

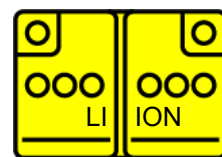
INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE FOR VEHICLE

OUTLANDER

for 48V MILD HYBRID

2026 Model year -



Version:1.0

Introduction

This manual provides safety instructions that need to be followed when rescuing the passengers from the vehicle after an accident and describes how to handle the damaged vehicle.

Please read and understand this manual carefully for your and the passengers safety.

Throughout this manual the words **WARNING AND CAUTION** appear. These serve as reminders to be especially careful. Failure to follow instructions could result in personal injury or damage to your vehicle.



WARNING

Indicates a strong possibility of severe personal injury or death if instructions are not followed.



CAUTION

Means hazards or unsafe practices that could cause minor personal injury or damage to the vehicle.



NOTE

Gives helpful information.

*: indicates optional equipment.

It may differ according to the sales classification; refer to the sales catalog.

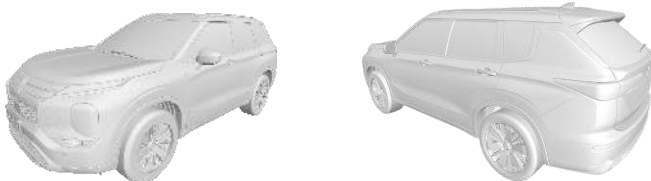
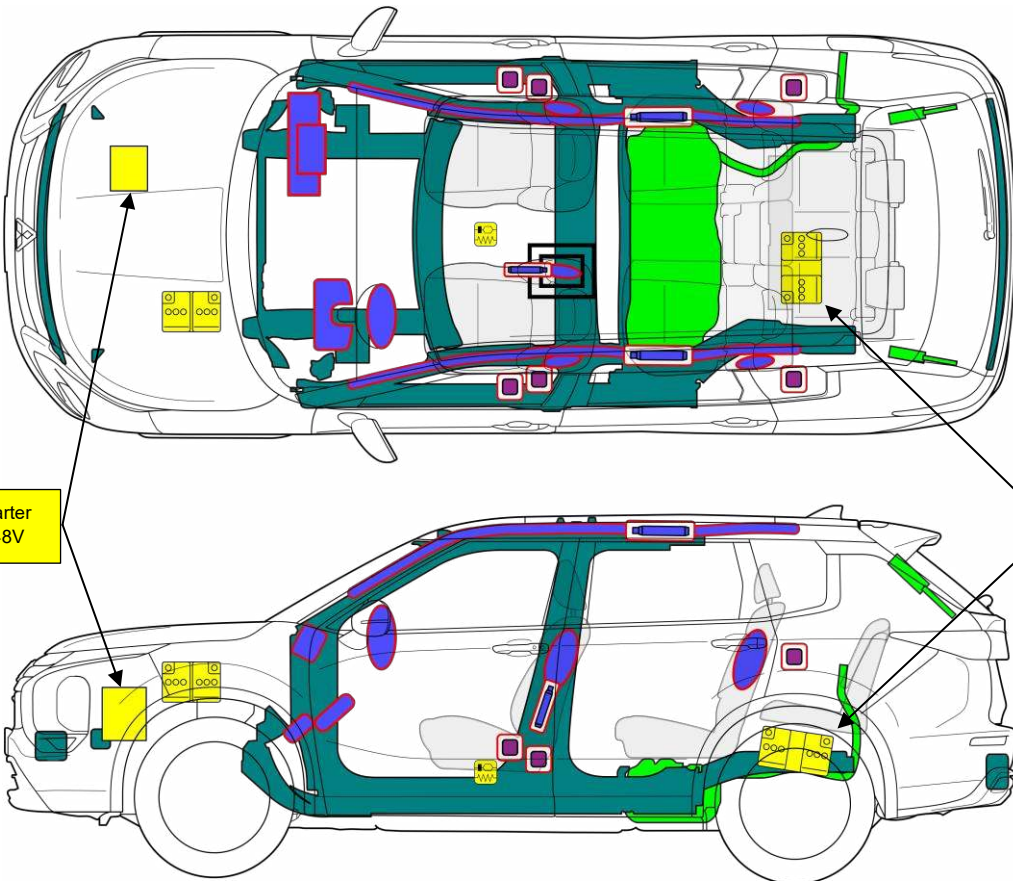
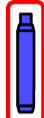
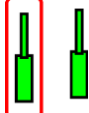
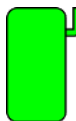
Mitsubishi Motors reserves the right to make changes in design and specification and/or to make additions to or improvements in this product without obligation to install them on products previously manufactured.

- Please note that the contents of this manual may differ with the actual vehicle due to vehicle specification changes.

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0. Rescue sheet(s)

 MITSUBISHI MOTORS		MITSUBISHI OUTLANDER 5 doors / 7 seats / Station wagon (2026 -)							
									
 Belt drive starter generator, 48V 48V Li-ion									
GM2W0001									
	Airbag		Stored gas inflator		Seat belt pretensioner		SRS control unit		Pedestrian protection active system
	Automatic rollover protection system		Gas strut / Preloaded spring		High strength zone		Zone requiring special attention		
	Battery low voltage		Ultra capacitor, low voltage		Fuel tank		Gas tank		Safety valve
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1. Identification / recognition

1. Features on vehicle exterior

The features of OUTLANDER are as follows:

WARNING:

If the rear cover of the belt drive starter generator is damaged while the engine is running, always wear insulated Personal Protective Equipment (PPE) (Rubber insulating gloves, Rubber soled insulating shoes).



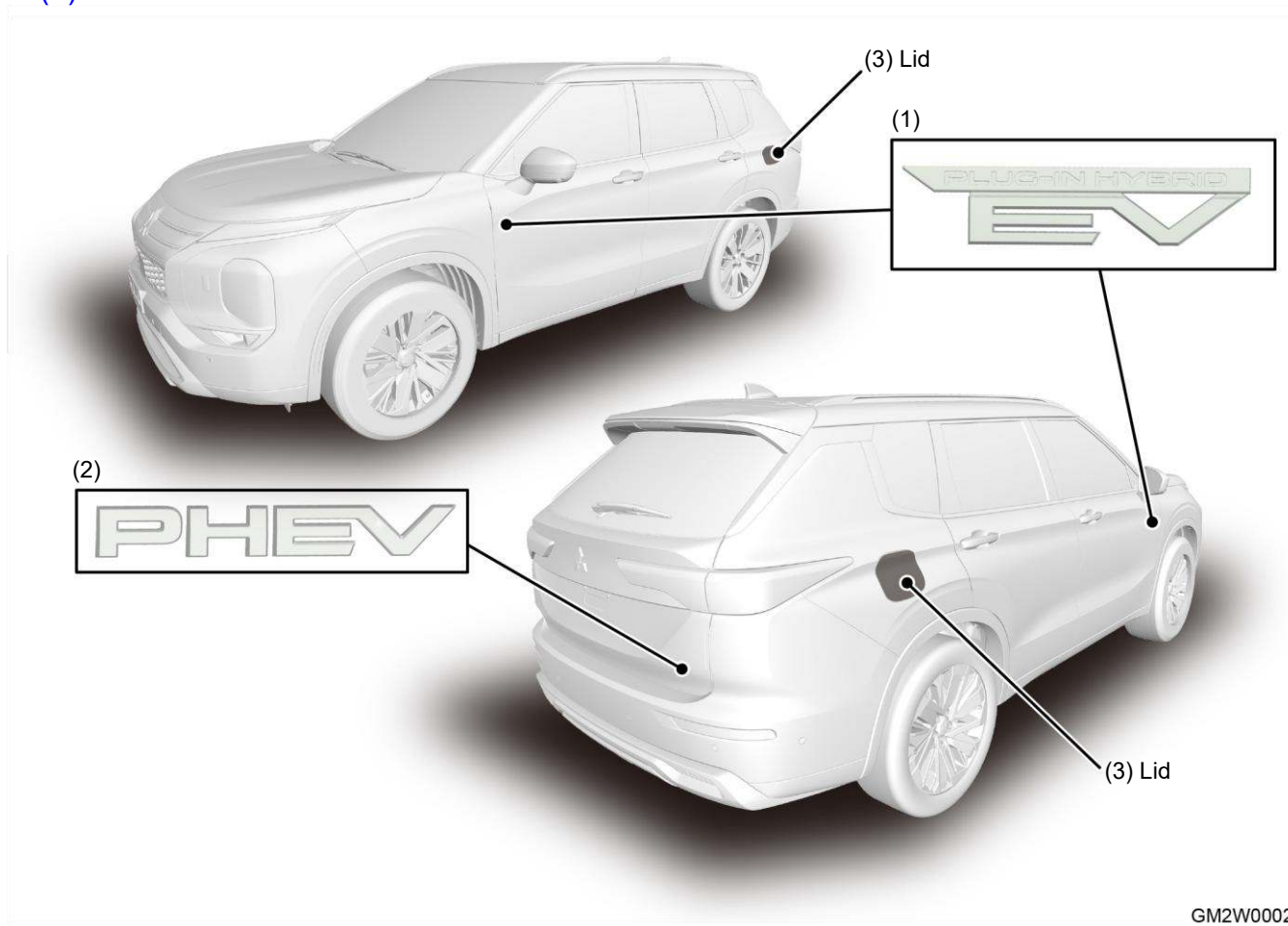
(1) Feature list

OUTLANDER can be identified by the logo, rear panel lid, and chassis number (model code).

Unlike OUTLANDER PHEV, OUTLANDER does not have the “PHEV (PLUG-IN HYBRID EV)” logo.

OUTLANDER has a lid only on the right rear panel, whereas OUTLANDER PHEV has lids on both rear panels.

(2) Features on exterior

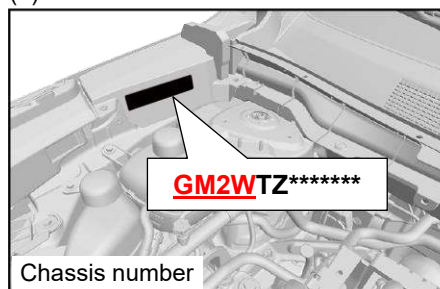


- (1) “PLUG-IN HYBRID EV” logo mark on the left and right sides of the door panel
- (2) “PHEV” logo mark on the liftgate
- (3) “Lid” on the rear panel

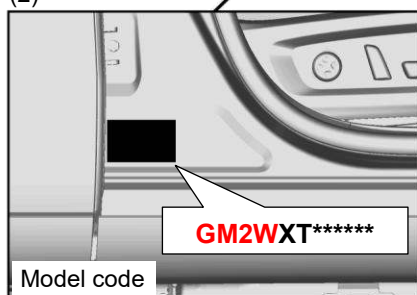
GM2W0002

(3) Features on other areas

(1)



(2)

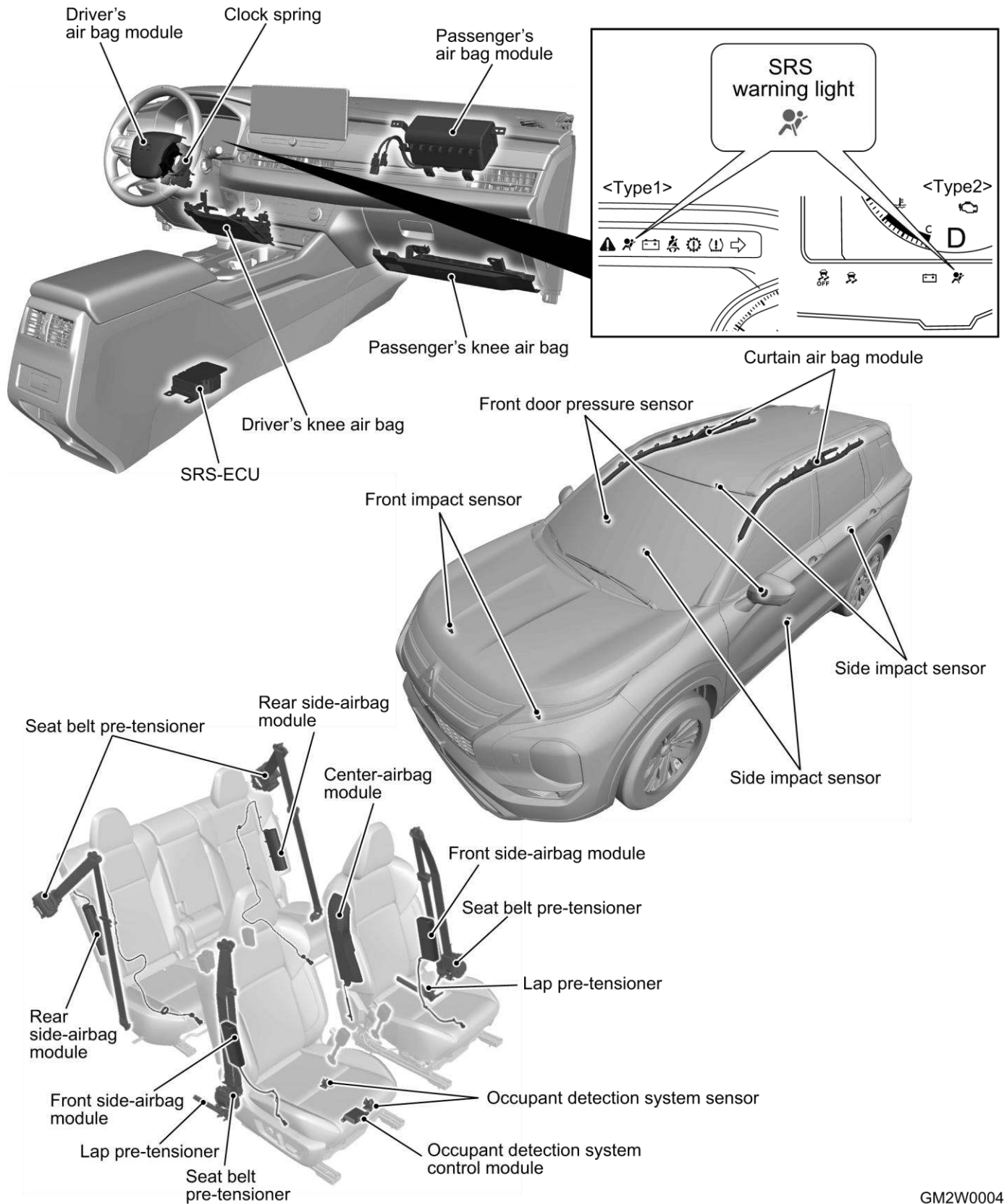


GM2W0003

- (1) "Chassis number" shown in the engine compartment
- (2) "Model code" shown in the Vehicle information code plate

2. SRS air bag component location

SRS air bags system (location of air bags and related components) are located as shown in the figure below.

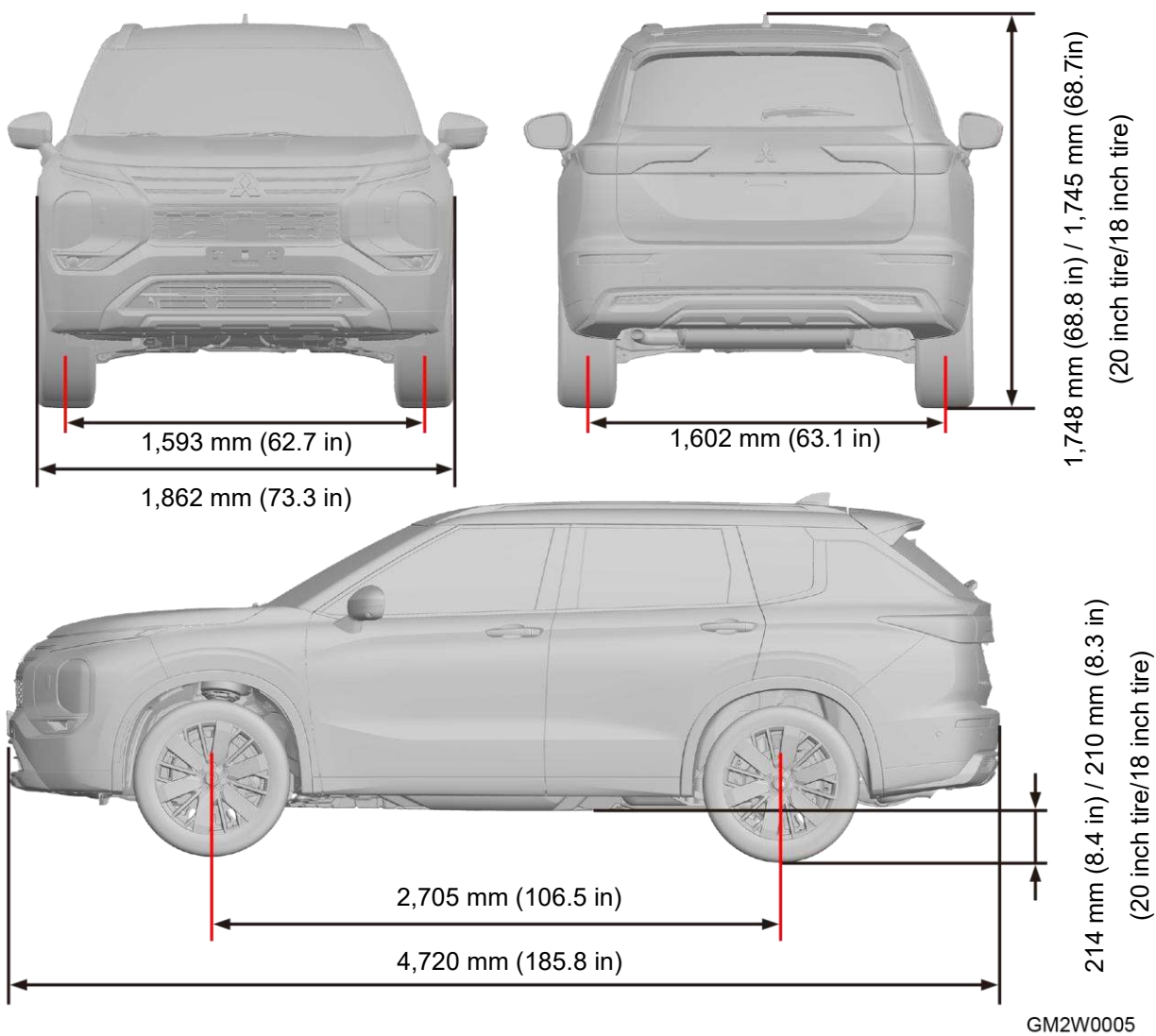


GM2W0004



*: Refer to "10. Explanation of pictograms used"

3. Vehicle dimensions



Vehicle weight: 3,770 – 4,043 lbs.*
1,710 – 1,834 kg*

* : Differs on the model, the vehicle weight will change.

2. Immobilization / stabilization / lifting

1. How to determine if vehicle is ON / OFF

The operating range of the keyless operation key is limited to the interior of the vehicle.

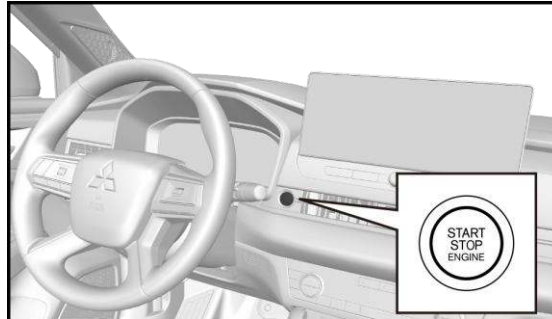


OFF

Check that the illumination of the "engine switch" is off.

ON

When the engine switch is pushed without depressing the brake pedal, the "engine switch" will illuminate.



GM2W0006

2. Support positions for Jack and Lift

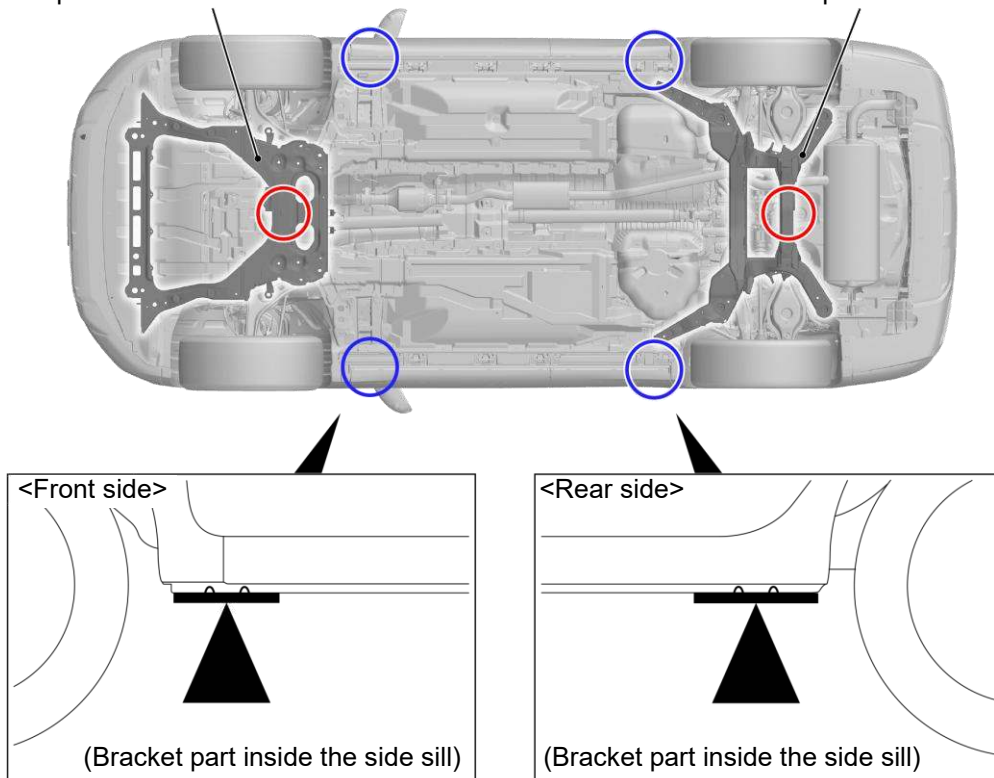


CAUTION:

Be sure to support the specified locations only. Otherwise, damage to the vehicle may occur.

Front suspension crossmember

Rear suspension crossmember



GM2W0007

○:Support position for Garage Jack ○:Support position for Jack or Axle stands or Lift

3. Vehicle Immobilization and Stabilization

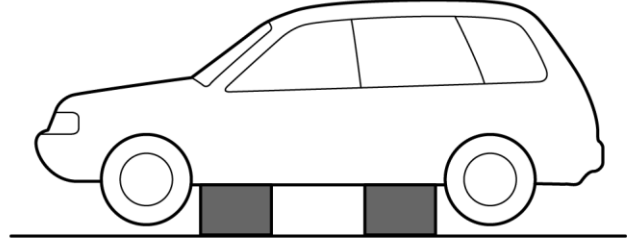
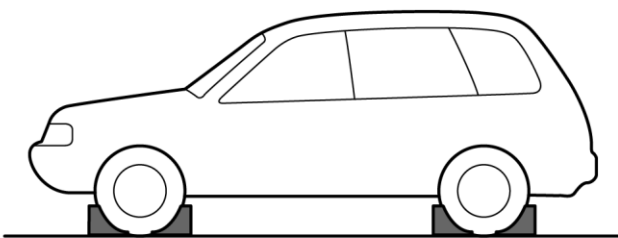
(1) Immobilization and stabilization method

⚠ WARNING:

- To apply the electric parking brake, press the brake pedal firmly to completely stop the vehicle and pull up the switch.
- When installing a “block” or “lift air bag device”, avoid exhaust systems and fuel systems, etc.

📌 NOTE:

When the 12V starter battery is weak or dead, or the battery terminal is disconnected, the electric parking brake cannot be applied or released.



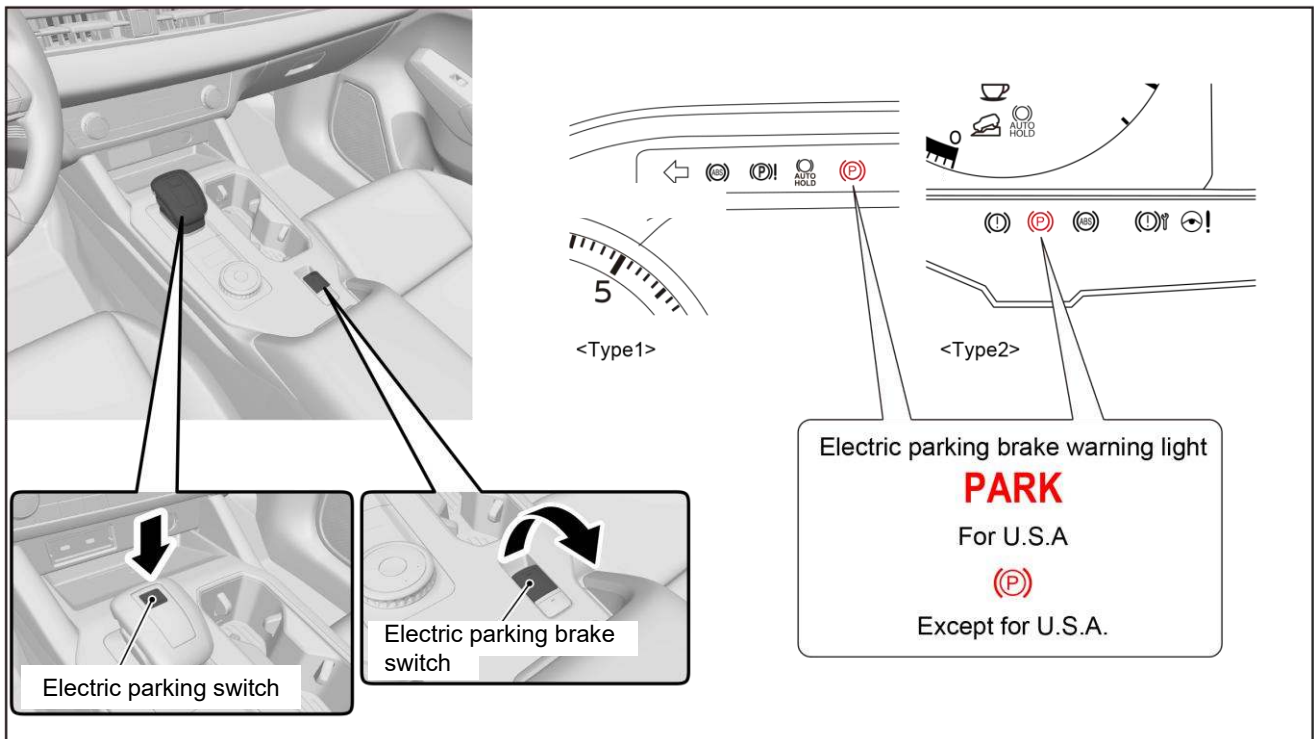
NE0040EEM0

• Immobilization with wheel chocks

Park the vehicle and press the electric parking switch.

Pull up the electric parking brake switch while depressing the brake pedal, check for the indicator light on the electric parking brake switch will come on.

Once the electric parking brake is applied, lock the vehicle with the wheel chocks.



GM2W0008

• Stabilize with block

To stabilize the vehicle, place a support such as a block for under the vehicle and release air from tires to stabilize the vehicle.

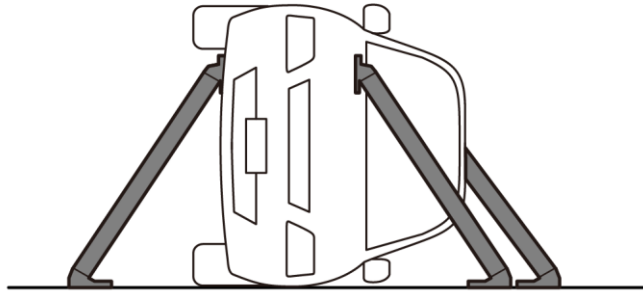
When fixing the vehicle to the foundation, use the designated position of the jack.

(2) Stabilization method for a rollover or overturned vehicle

⚠ WARNING:

- Use supports with sufficient strength.
- When installing prop tools, avoid exhaust systems, fuel systems and other areas where use of the tool could cause personal injuries or damage to the vehicle.

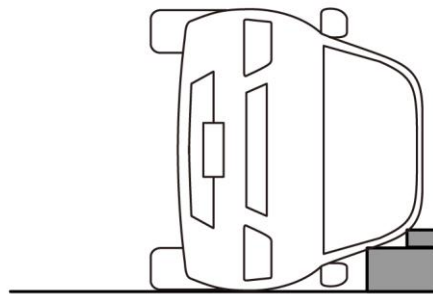
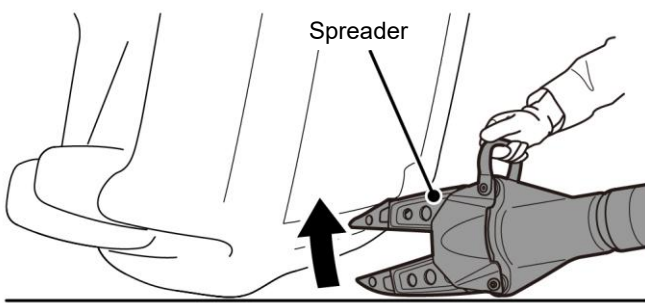
• Stabilization method for rollover vehicle using prop tools.



NE0073EJM0

The prop tools should be placed on the vehicle roof or the vehicle bottom, as depicted above. Depending on the situation, use multiple prop tools to stabilize the vehicle.

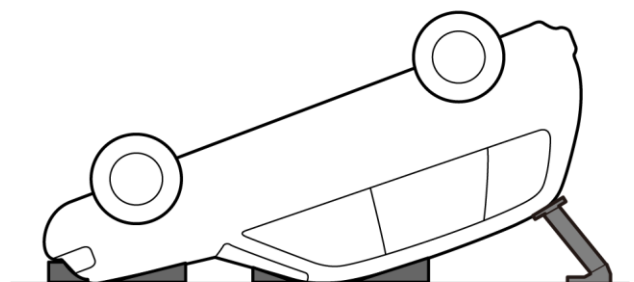
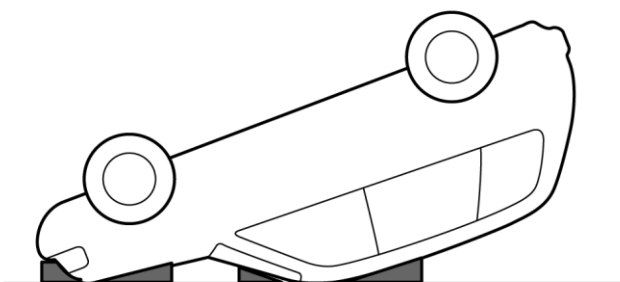
• Stabilization method for rollover vehicle using blocks.



NE0072EJM0

To upright the vehicle, use a spreader or jack, etc., and place supports such as blocks in the gap to stabilize it.

• Stabilization method for overturned vehicle using wheel stoppers, blocks, and prop tools.



NE0076EJM0

Blocks or wheel stoppers are set up in the gap between the hood and pillars to stabilize the vehicle. To increase the stability, place a prop tool at the rear of the vehicle to stabilize it.

3. Disable direct hazards / safety regulations



NOTE:

Disconnecting the 12V starter battery may make it impossible to perform operations related to electrical components. Perform the necessary operations before removing the 12V starter battery terminal.

OUTLANDER is equipped with a belt drive starter generator instead of an alternator.

The drive starter generator is directly connected to the crankshaft pulley via a drive belt, and the vehicle does not operate in a mode where only the belt drive starter generator runs while the engine is stopped.



CAUTION:

- (1) Incorrect handling of the vehicle may result in electric shock or other serious injuries. Please exercise extreme caution.
- (2) To reduce the risk of injury or electric shock, follow the procedures described in this manual carefully.
- (3) On this vehicle, when the engine is running, certain conductive parts inside the belt drive starter generator cover could be energized with voltages momentarily exceeding AC 30 V.



WARNING:

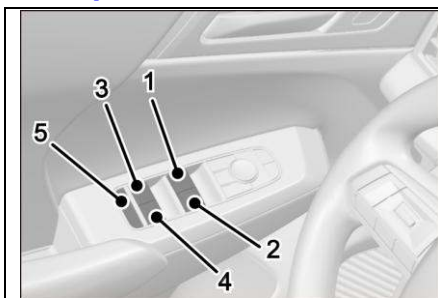
If the rear cover of the belt drive starter generator is damaged while the engine is running, always wear insulated Personal Protective Equipment (PPE) (Rubber insulating gloves, Rubber soled insulating shoes).



NOTE:

Disconnecting the 12V starter battery may make it impossible to perform operations related to electrical components. Perform the necessary operations before removing the 12V starter battery terminal.

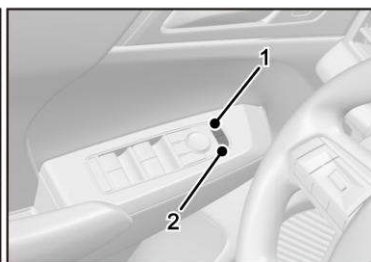
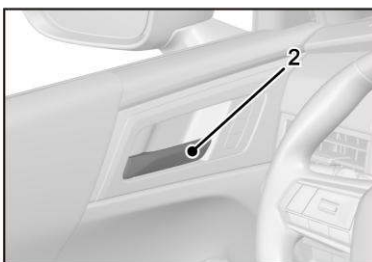
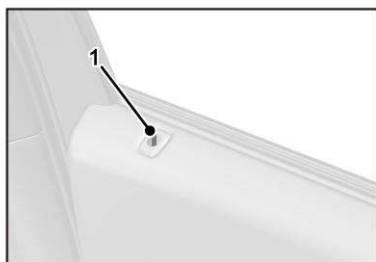
1. Electric power windows / Door lock



GM2W0009

Power windows

- 1: Front left door window
- 2: Front right door window
- 3: Rear left door window
- 4: Rear right door window
- 5: Window lock switch



GM2W0010

Door lock

- 1 – Lock
- 2 – Unlock

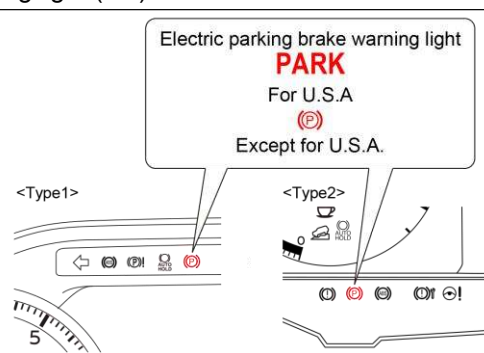
NOTE:

When operating the switch on the armrest, all the doors are locked/unlocked together. Pull the handle twice when unlocking the door using the handle other than driver's side one. When opening the driver's side door, all the doors are unlocked.

2. How to apply and release the parking brake

•To apply

- 1) Stop the vehicle completely.
- 2) Pull up the electric parking brake switch while depressing the brake pedal.
- 3) When the parking brake is applied, the electric parking brake warning light (red) in the instrument cluster will come on.



GM2W0011

Make sure that the electric parking brake warning light (red) in the combination meter and the indicator light in the switch are lit.

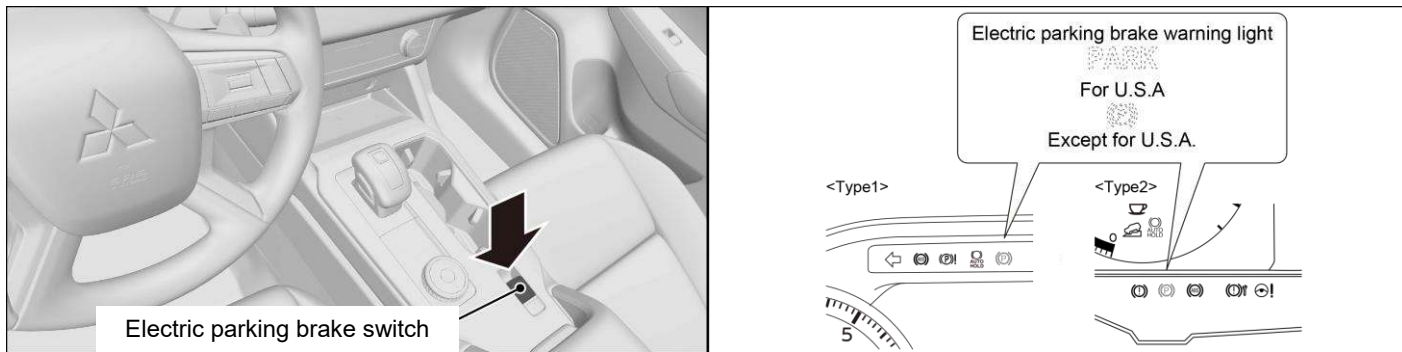
CAUTION:

When applying the electric parking brake, stepping on the brake pedal firmly, stop completely the vehicle, and then pull up the switch.

•To release

Manual operation

- 1) Turn the power supply mode of the engine switch to the ON position.
 - 2) Press down the electric parking brake switch while depressing the brake pedal.
- When the parking brake is released, the electric parking brake warning light (red) go off.



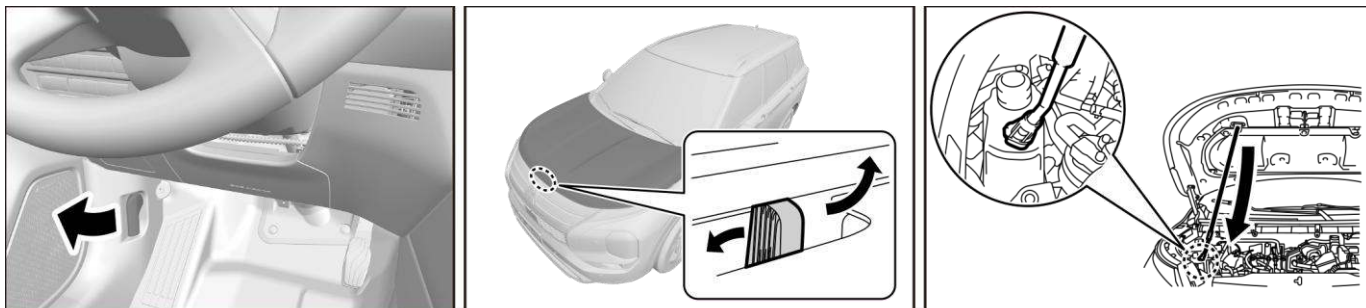
Automatic operation

When the accelerator pedal is depressed slowly while all of the following conditions are met, the electric parking brake is automatically released.

- The engine is running.
- The select position is in the "D" (DRIVE) or the "R" (REVERSE) position.
- The driver's door is closed.

3. How to open hood

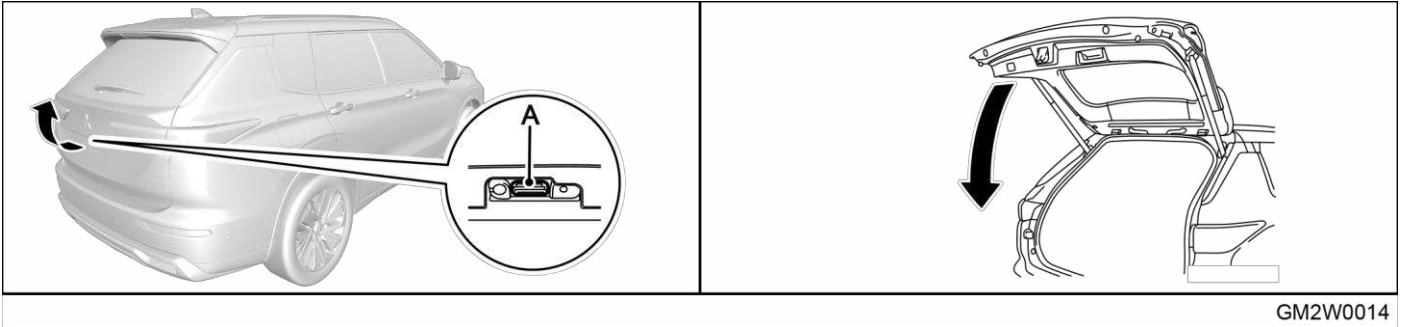
- 1) Pull the release lever towards you to unlock the hood.
- 2) Raise the hood while pressing the safety lock.
- 3) Support the hood by inserting the support bar in its slot.



4. How to open liftgate

<Manual liftgate>

Unlocking the liftgate, push the liftgate opener switch (A) and pull up the liftgate.



<Power liftgate>

• Operating the power liftgate using the inside power liftgate switch

Pushing the power liftgate switch (A) on more than 1 second.

If the switch is pushed while the power liftgate is being opened or closed, the power liftgate will stop.

Pressing the power liftgate switch (A) again to open or closed.

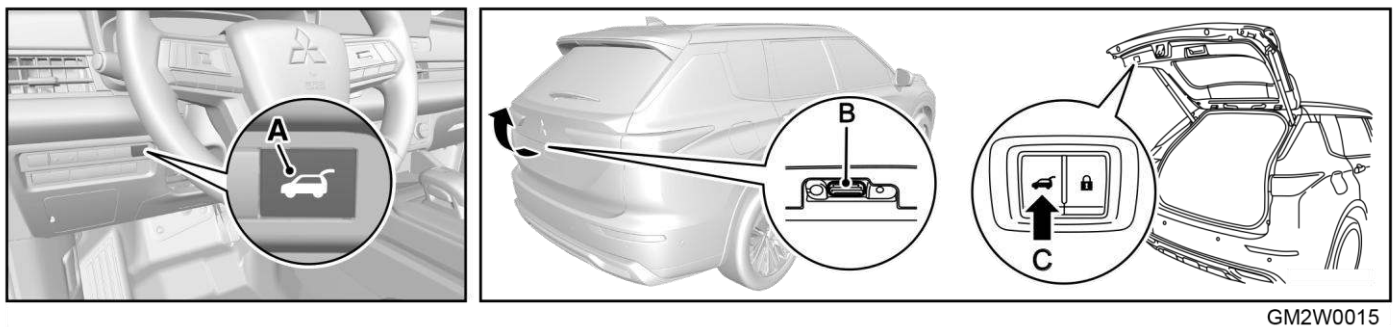
• Opening the power liftgate using the liftgate opener switch

Power liftgate can be auto opened by pressing the liftgate opener switch (B).

The power liftgate can be closed by pressing the power liftgate close switch (C) on the inside of the power liftgate.

If the switch is pushed while the liftgate is being opened or closed, the liftgate will stop.

Pressing the liftgate opener switch (B) or power liftgate close switch (C) again to open or close.



• Operating the power liftgate using the keyless operation key

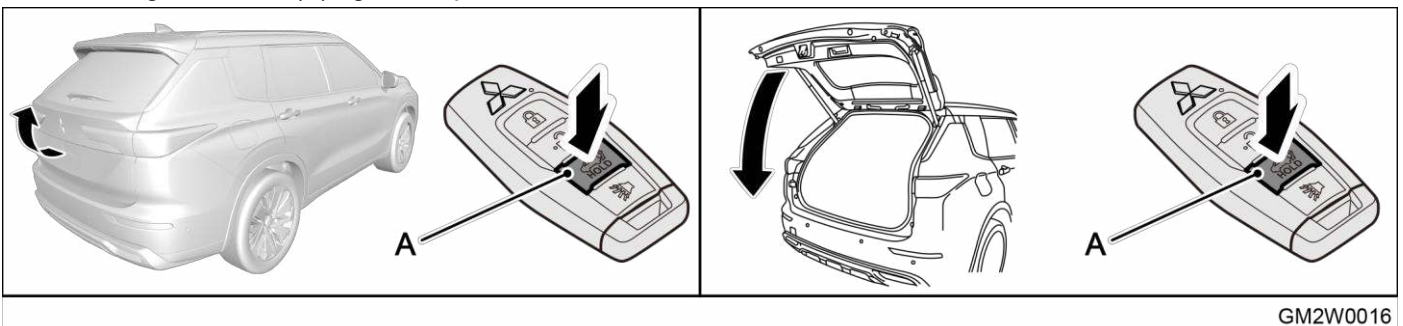
Pushing the power liftgate button (A) on more than 1 second.

- While closed: The liftgate completely opens.

- While open: The liftgate completely closes.

If the button is pushed while the liftgate is being opened or closed, the liftgate will stop.

Pressing the button (A) again to open or close.



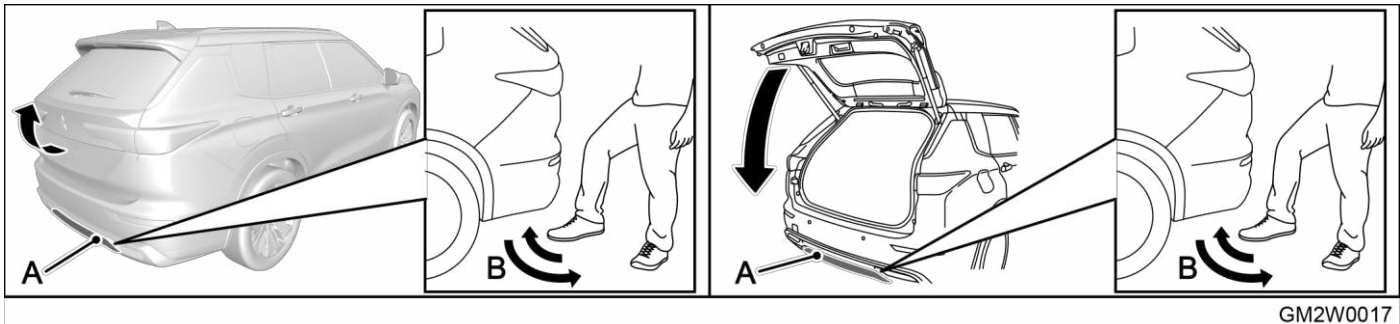
• Operating the power liftgate using the hands-free access

NOTE:

The kick motion sensor will not respond unless you carry a keyless operation key.

The kick motion sensor (A), located on the back of the rear bumper, enables you to open or close the liftgate in hands-free.

When you move your foot under and away from the operating range (B) similarly to a kicking motion, the liftgate will open or close automatically.



GM2W0017

The power liftgate will fully open automatically using the kick motion sensor (A).

- 1) Carry the keyless operation key.
- 2) Move your foot under and away from the rear bumper similarly to a kicking motion within the operation range of the kick motion sensor.
- 3) The liftgate will automatically open or close.
 - While closed: The liftgate completely opens.
 - While open: The liftgate completely closes.

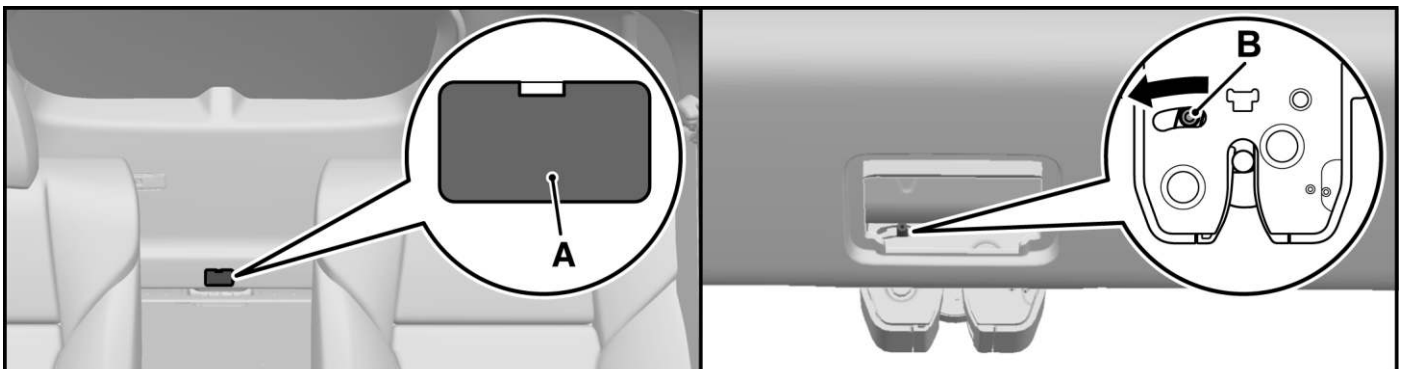
If the kick motion while the liftgate is being opened or closed, the liftgate will stop.

It can be opened and closed again with kick motion.

•To open from inside

The inside liftgate release is designed to provide a way to open the liftgate in the case of a discharged 12V starter battery.

- 1) Open the lid (A) inside of the liftgate.
- 2) Using a tool, move the lever (B) to open the liftgate.



DE0049ENM0

5. How to disconnect the 12V starter battery negative terminal

Use an open end wrench [10 mm (0.4 in)] or ratchets wrench [10 mm (0.4 in)] to disconnect the negative terminal of the 12V starter battery according to the procedure below, and then wrap a plastic tape around the disconnected negative terminal.

Shut down the SRS air bag system circuit by disconnecting the negative terminal of the 12V starter battery.

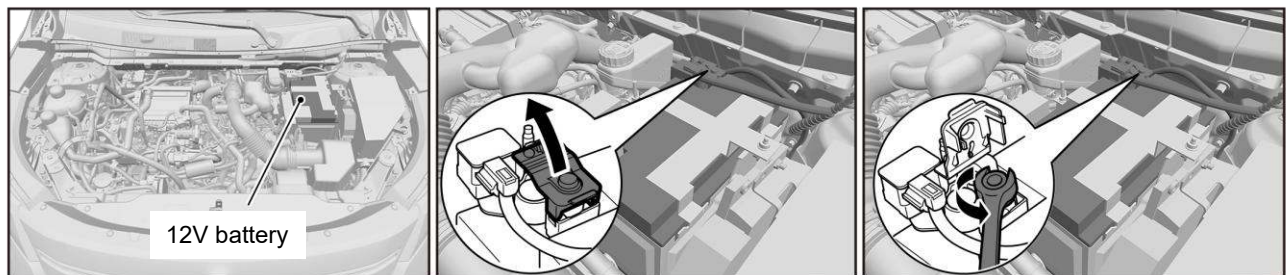
- 1) Open the hood and keep it open.
- 2) Turn the power supply mode of the engine switch to the ON position.
- 3) Turn the power supply mode of the engine switch to the OFF position with the driver side door opened.

CAUTION:

- After closing the driver's door, do not disconnect the negative terminal of the 12V starter battery for 3 minutes.
- While waiting, never operate the vehicle such as locking, opening, and closing doors.



- 4) Close the driver's door and wait at least 3 minutes.
- 5) Check that the P indicator in the combination meter and the electric parking switch is off.
- 6) Pull up the lid cover of the battery current sensor.
- 7) Disconnect the 12V starter battery negative terminal.



GM2W0018

4. Access to the occupants

Failure to follow these instructions when performing a rescue may result in serious injury such as electric shock.



NOTE:

Disconnecting the 12V starter battery may make it impossible to perform operations related to power seat. Perform the necessary operations before removing the 12V starter battery terminal.



WARNING:

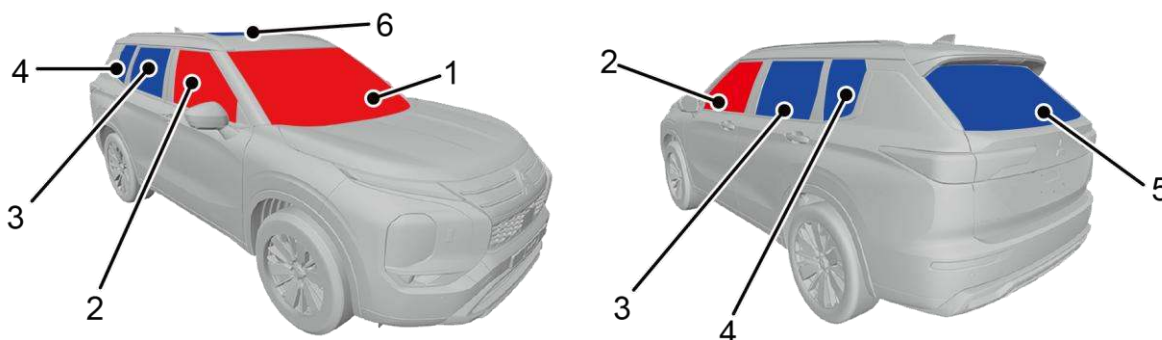
If the rear cover of the belt drive starter generator is damaged while the engine is running, always wear insulated Personal Protective Equipment (PPE) (Rubber insulating gloves, Rubber soled insulating shoes).



CAUTION:

- If the 12-volt starter battery is disconnected or removed, the power liftgate system will not operate. Do not close the liftgate in this condition, as it cannot be reopened until battery power is restored.
- The electric parking brake system will also be inoperative when the 12V starter battery is disconnected.

1. Windows



GM2W0019

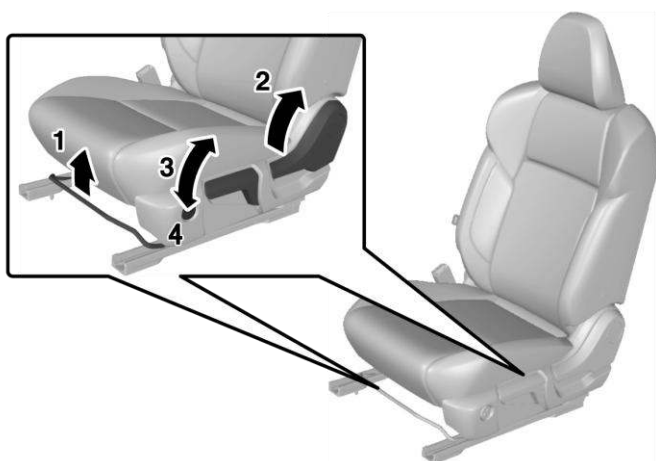
1 – 2 : Laminated glass

3 – 6 : Tempered glass

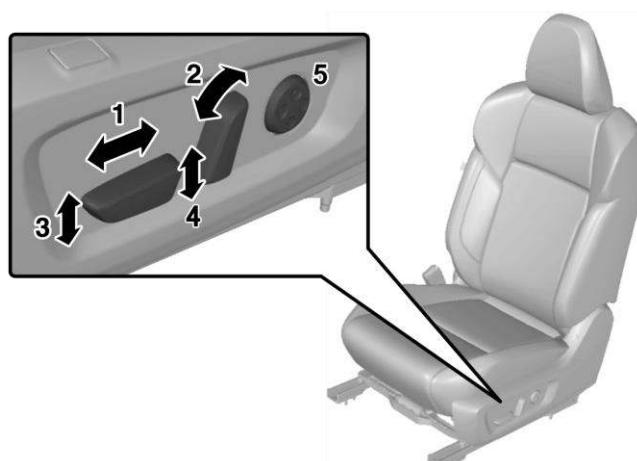
2. Adjustment seat and steering wheel

-Seat

< Manual seat type >



< Power seat type >



DE0021ENM0

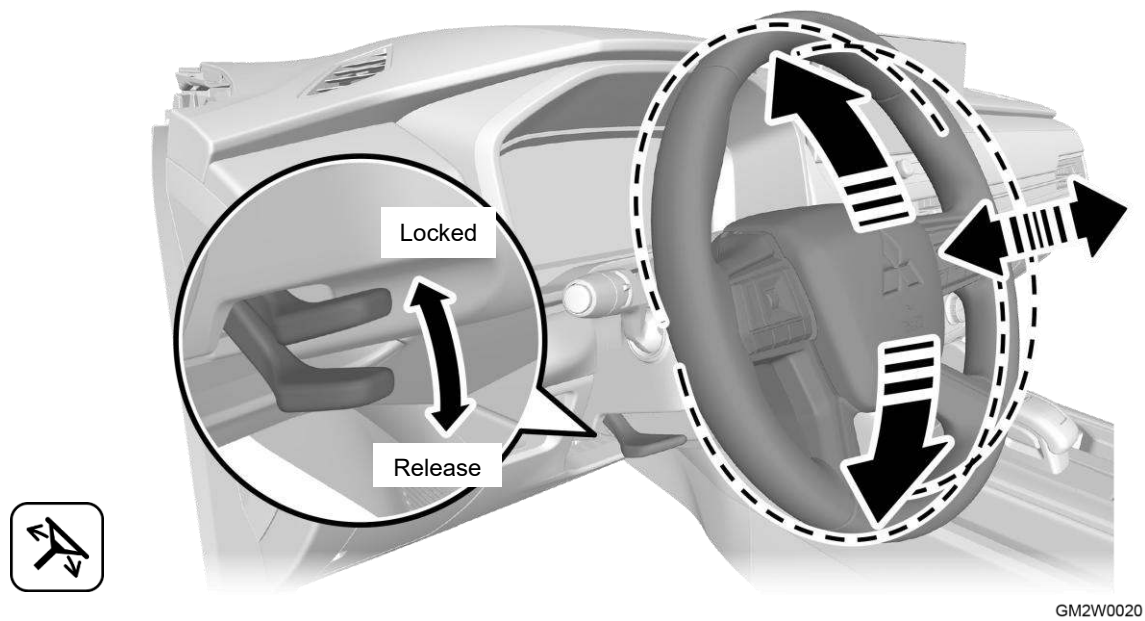
- 1 : To adjust forward or backward
2 : To recline the seatback
3 : To adjust seat height (driver's side only)
4 : To adjust lumbar support (driver's side only)

- 1 : To adjust forward or backward
2 : To recline the seatback
3 : To adjust seat cushion angle (driver's side only)
4 : To adjust seat height (driver's side only)
5 : To adjust lumbar support (driver's side only)



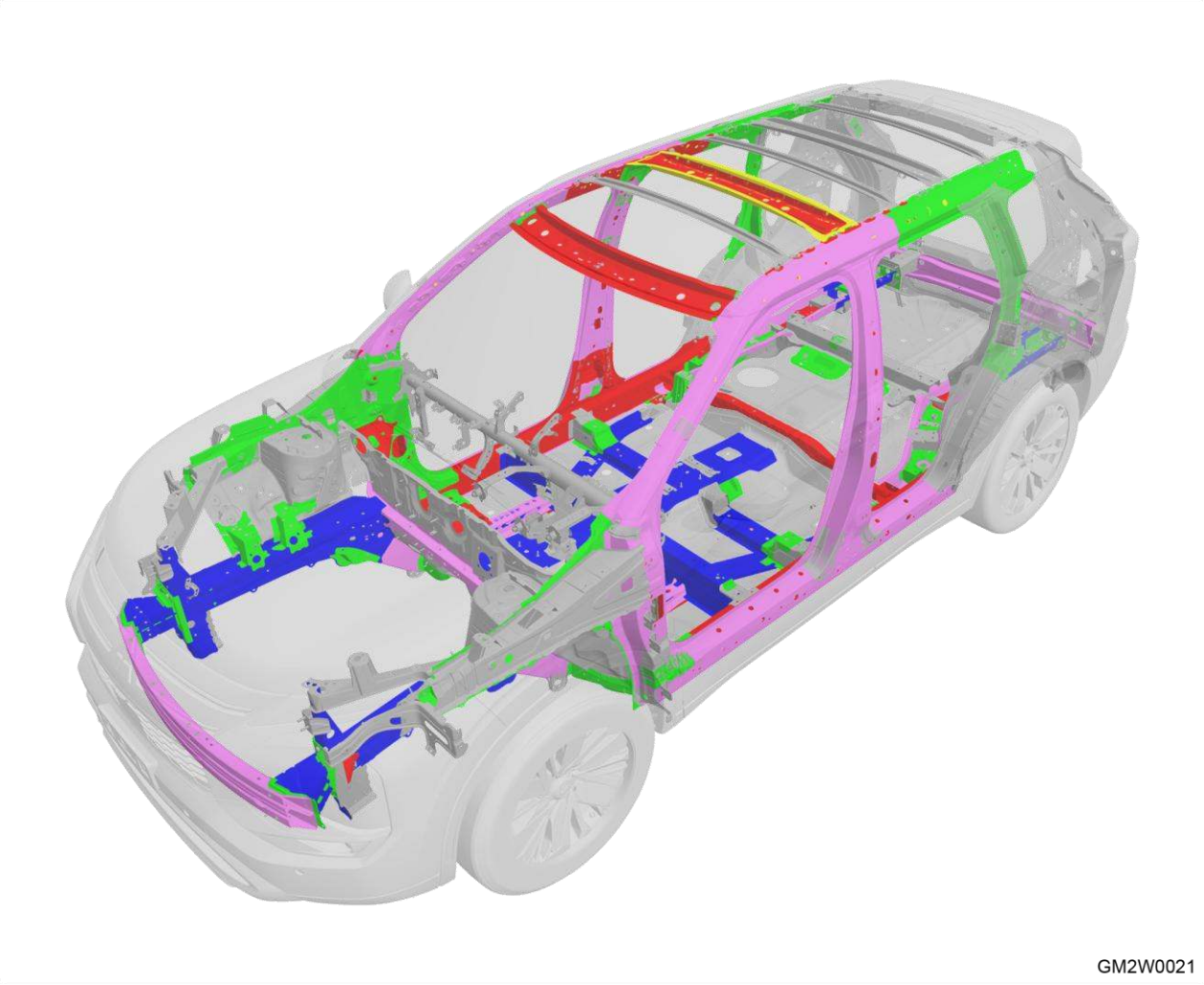
*: Refer to "10. Explanation of pictograms used"

•Steering wheel



*: Refer to “10. Explanation of pictograms used”

3. High-tensile and Ultra-high-tensile steel panels location



 : - 440 MPa	 : 590 MPa	 : 780 MPa	 : 980 MPa
 : 1180 MPa	 : 1470 MPa		

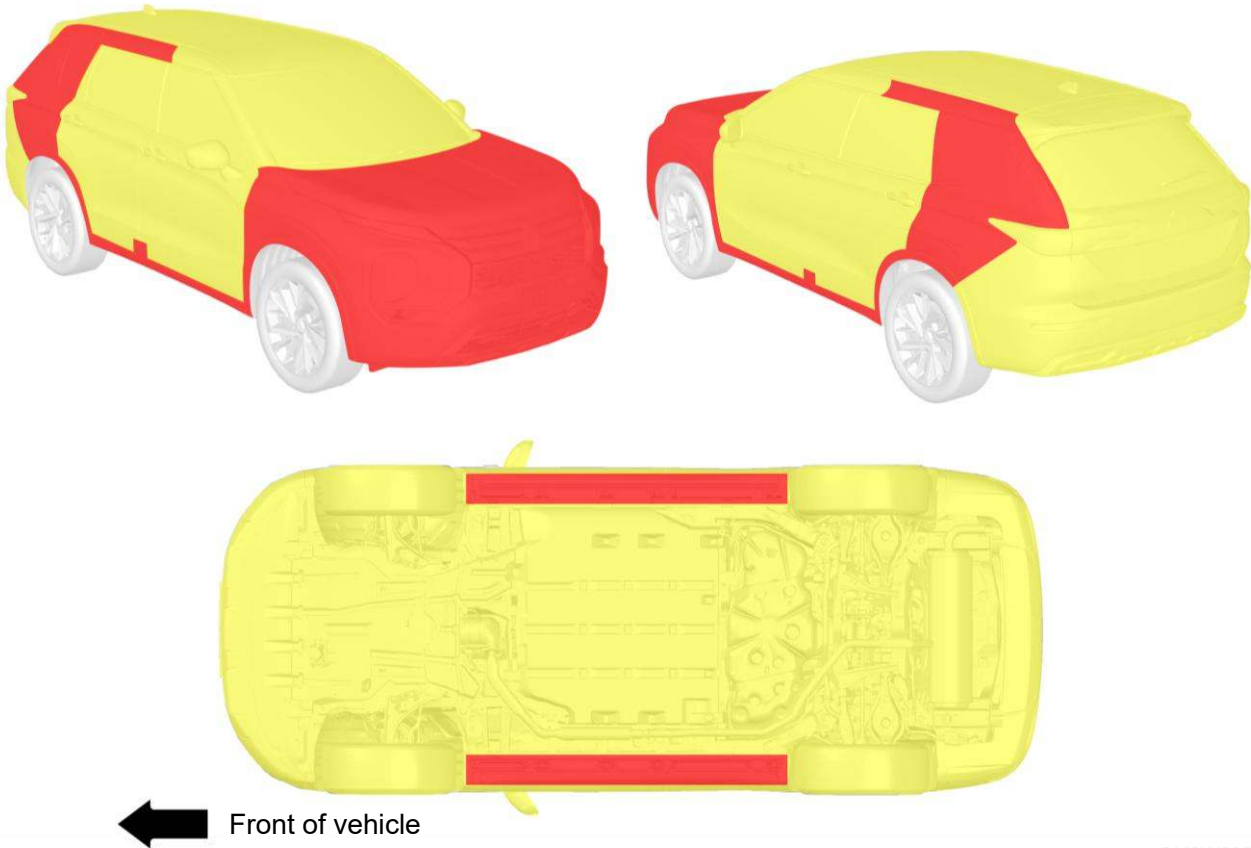
4. Vehicle CUT Zones

If you need to cut the vehicle body, cut the vehicle body and perform rescue work.

**WARNING:**

- Use a hydraulic cutter or a suitable tool that does not generate sparks to cut the vehicle body. Failure to do so could result in serious personal injuries to you and occupants of the vehicle.

	<u>Risk of air bag deployment</u> Do not cut this area because there is risk that an air bag may be deployed due to a short circuit or an impact caused by the accident. If all air bags have deployed, this area can be cut. If at least one minute has elapsed after disconnecting the negative terminal of 12V starter battery or changing the power supply mode of the engine switch to OFF, this area can be also cut.
	<u>CUT ZONE</u>



Front of vehicle

Bottom view

GM2W0022

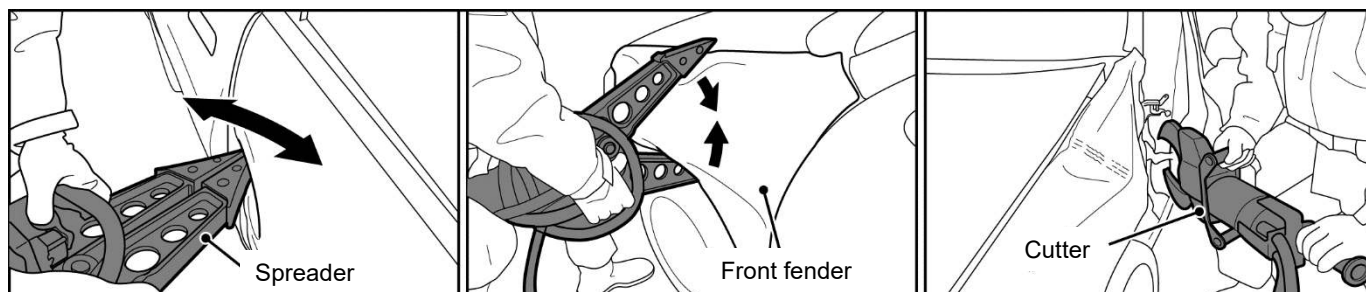
5. Opening the door with a spreader or cutter

⚠ WARNING:

- Use a hydraulic cutter or a suitable tool that does not generate sparks to cut the vehicle body. Failure to do so could result in serious personal injuries to you and occupants of the vehicle.

If you cannot open the door from the outside, use a spreader, cutter, or similar tool to rescue the door if necessary.

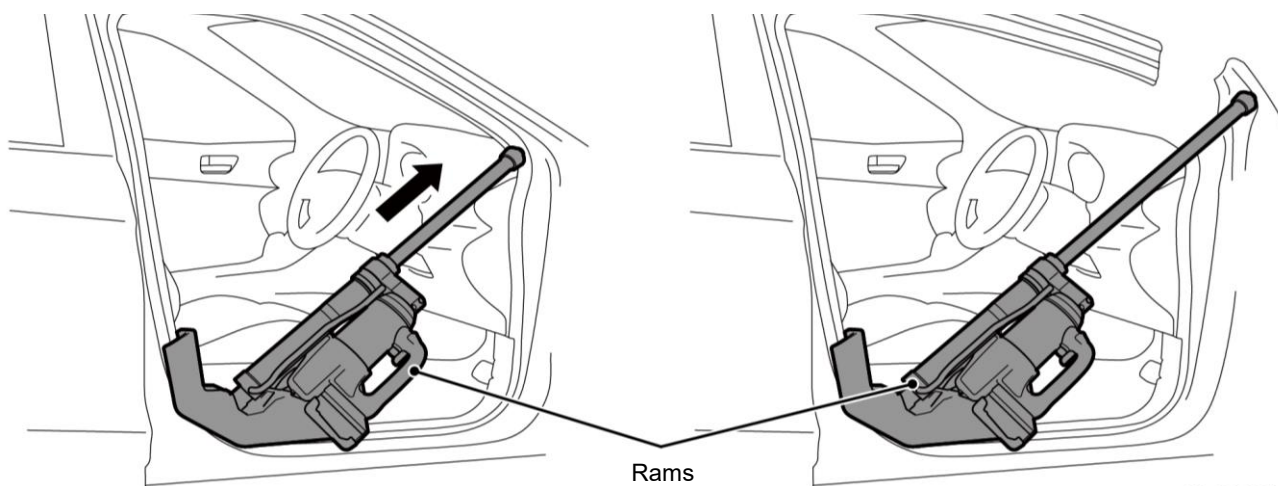
- 1) Insert the tip of the spreader into the door latch and open the spreader to make a gap.
- 2) If necessary, compress the front fender with a spreader to make a gap in the door hinge part.
- 3) Insert the cutter into the gap between the door hinges created, to break it hinges and open the door.



NE0074EJM0

6. Securing space by rams

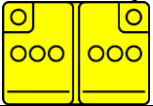
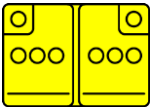
If the body is crushed and you cannot rescue the occupant, use a rams on the crushed part to push the body wide and secure a space for rescue. If necessary, cut the pillar part and expand it.



NE0075EJM0

5. Stored energy / liquids / gases / solids

Fluids / gases used in this vehicle

	Capacity & Type	Color
Fuel tank (Petrol) 	55 liters (14.5 gallons)	Pale amber (87 Octane rating)
48V battery 	Total voltage: 44V Total electric energy: 370 Wh	Clear & colorless
12V starter battery 	12V – L3-EFB (70 Ah / 720 A)	Clear & colorless
Engine oil	4.3 liters (4.54 quarts)	Dark brown
Engine coolant	8.0 liters (8.45 quarts)	Blue
Intercooler coolant	3.2 liters (3.28 quarts)	Blue
Brake fluid	As required	Clear or yellow or brown
CVT fluid	8.6 liters (9.09 quarts)	Red
Differential gear oil	0.5 liters (0.53 quarts)	Dark brown
Transfer fluid	0.3 liters (0.33 quarts)	Dark brown
Coupling oil	0.6 liters (0.63 quarts)	Dark brown
Refrigerant (air conditioning) 	HFO-1234yf: 550 g (19.4 oz)	Non color

WARNING:

A flammable carbonic acid ester solution containing lithium salt is used as the electrolyte for the 48V battery.



*: Refer to “10. Explanation of pictograms used”

About flammable substances

Flammable substances	Flame-retardant substance
• Plastic • Electrolyte substance • Oil / Gasoline • Flammable gas * • Refrigerant gas: HFO-1234yf (R-1234yf) • Others	• Carbon fiber • Refrigerant gas: HFC-134a (R-134a)

*: Combustible gas generated when the 48V battery is deformed or damaged.

6. In case of fire

1. Vehicle fire

In case of vehicle fire, contact the fire department immediately and follow the steps below to start extinguishing the fire.

In the event of a vehicle fire or a 48V battery fire, use a large amount of water or an ABC powder fire extinguisher to extinguish and cool the fire.

In the case of a fire from the 48V battery part, fire or smoke will be emitted from the rear of the vehicle.

WARNING:

Failure to follow these instructions may result in serious injury such as electric shock:

- 1) The 48V battery is designed to prevent electrolyte leaking the 48V battery just in case it is broken.
- 2) The 48V battery uses an electrolyte made of flammable Carbonate ester solution of lithium salts.

When reacting with moisture in the air, this electrolyte generates acidic organic vapor which is harmful to human body.

- 3) Therefore, use appropriate Personal Protective Equipment (PPE) including mask for organic gas, solvent resistance gloves and eye protector and use appropriate caution.



*: Refer to “10. Explanation of pictograms used”

2. Fire-extinguishing



Use ABC powder fire extinguisher or large quantity water to extinguish the fire

*: Refer to “10. Explanation of pictograms used”

WARNING:

- Never use seawater or any water containing salt.
- Please note the following safety distances when extinguishing a fire:

When extinguishing a fire by spraying with water: Keep a distance of 1 m (3.3 ft) or more and extinguish the fire.

When the fire is extinguished when it is fully opened by discharging water (straight): Keep a distance of 5 m (16.4 ft) or more and extinguish the fire.

1) **By using fire extinguisher**

Use a fire extinguisher which is suitable for flammable liquid and electrical equipment fires.

2) **By using water**

Use water that does not contain salt, such as tap water, well water or pond water.

DO NOT attempt to extinguish the fire with a small amount of water as it is dangerous.

Use large quantity of water.

A large volume of water, such as from a fire hydrant must be used. Unless a large volume of salt-free water is available, keep away from the vehicle fire and wait for fire department to arrive.

CAUTION:

When there is a possibility that the water has gotten inside the 48V battery, treat the vehicle as a submerged vehicle. (Refer to “7. In case of submersion”)

7. In case of submersion

1. Submerged Vehicle

CAUTION:

If the 48V battery is immersed in water, they will be treated as submerged.

If the vehicle is submerged or partially submerged, remove occupants from the vehicle and water.

Then shut down the 12V starter battery.

If the vehicle is submerged, water may enter the 48V battery.

If the vehicle is submerged, the 48V battery may generate flammable hydrogen gas.

WARNING:

- If water enters the 48V battery, hydrogen gas may be generated.
- When seawater enters, a large amount of hydrogen gas is generated by rapid electrolysis due to salinity, which may cause a fire.
- After the vehicle is extracted from the water, confirm that no open flames or ignition sources are present. Once safety is ensured, open the windows and doors to ventilate any accumulated hydrogen gas.



GM2W0023

Rescue operation

If the vehicle is submerged or partially submerged, take the following rescue operation steps:

- 1) Rescue the occupants and pull the vehicle out of the water.
- 2) Inspect the vehicle for damage. Wear insulated Personal Protective Equipment (PPE) and carry out the rescue operation while taking care not to touch the 48V battery.
 - When the vehicle is severely damaged.
 - When the 48V battery is deformed or damaged, and internal parts are exposed.
 - When the damage condition of the 48V battery cannot be determined.

CAUTION:

If the 48V battery is damaged, consult the nearest certified Mitsubishi Dealer how to handle the 48V battery.

8. Towing / transportation / storage

1. How to Transport

Transport the vehicle on a flatbed truck or tow the vehicle with all wheels off ground

- Vehicle dimensions. ([Refer to 1-4.](#))



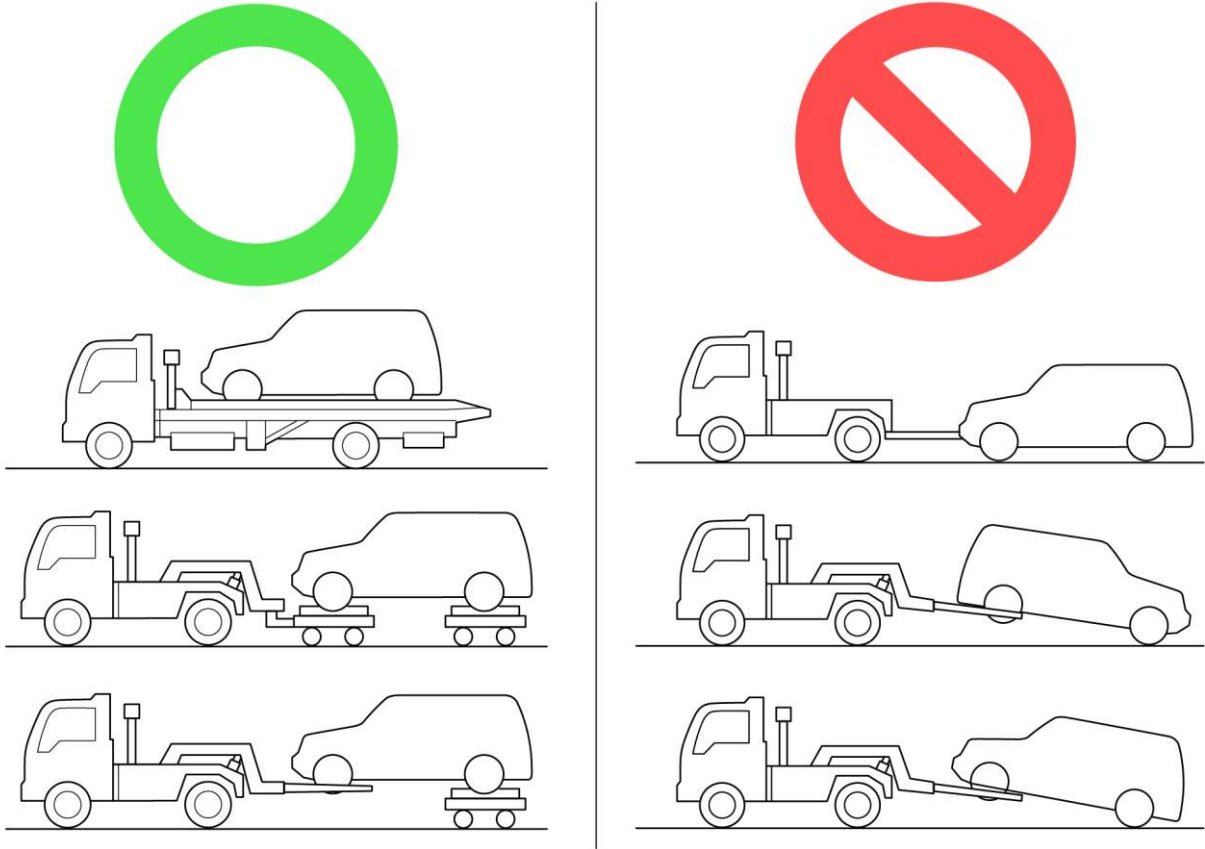
WARNING:

- Never tow the vehicle with front wheels and/or rear wheels on the ground. This may cause damage to the transaxle.



CAUTION:

If the 12V starter battery charging level is too low or the 12V starter battery negative terminal is disconnected, you cannot move the selector position from the P position.



NE0024EEM0

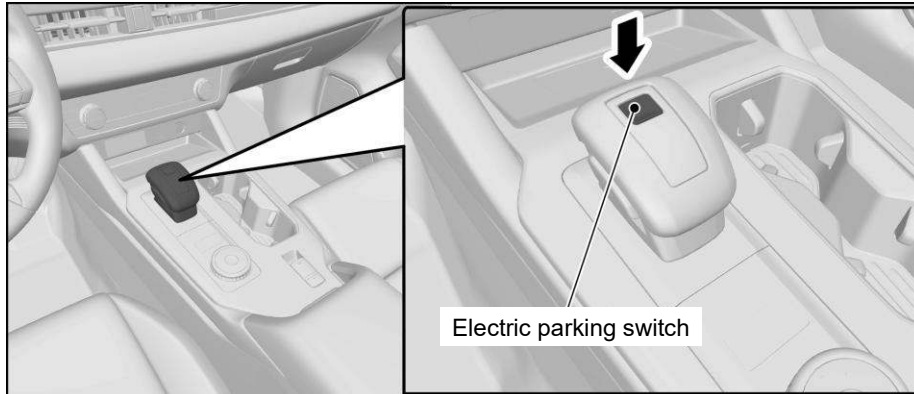
- The figure shows examples only.
- When loading the vehicle on the truck, handle carefully to prevent further damage.

2. How to release the electric parking brake manually (if required)

WARNING:

Releasing the electric parking brake manually should only be done in an emergency.

- 1) With the vehicle parked, push the electric parking switch, then chock the wheels.



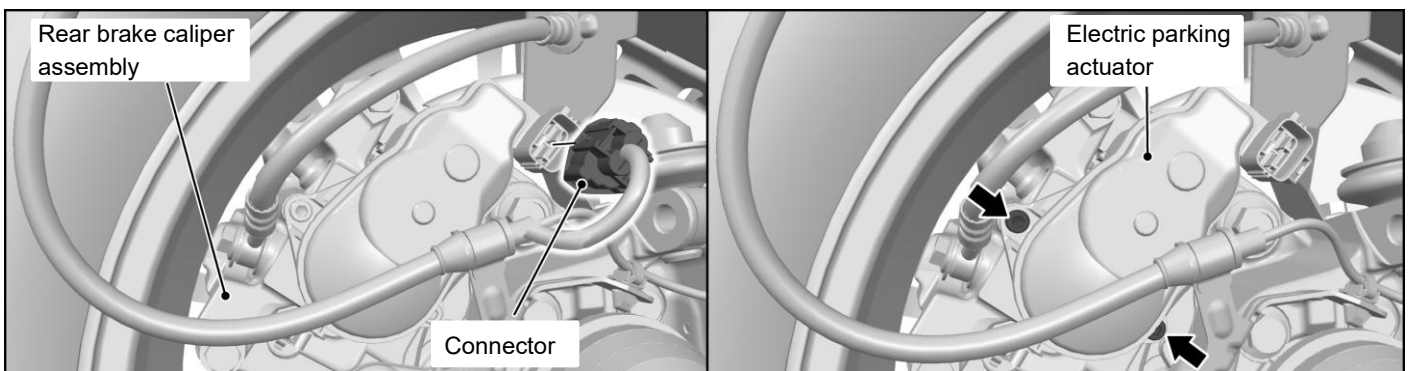
GM2W0034

- 2) Disconnect the 12V starter battery negative terminal. ([Refer to 3-5.](#))

CAUTION:

- If signs of dirt or water are found around the connector, clean it thoroughly.
- Ensure that no foreign materials intrude into the connector.

- 3) Disconnect the wiring harness connector from the electric parking actuator.
- 4) Remove the two bolts to remove the electric parking actuator from the rear brake caliper assembly.



DE0028ENM0

- 5) Insert a hexagonal wrench [6 mm (0.2 in)] to the rear brake caliper assembly, and then turn the wrench clockwise 2 turns



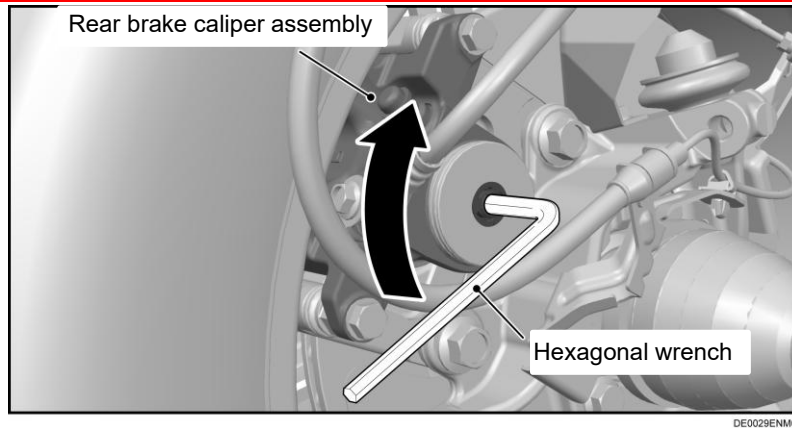
WARNING:

Depress the brake pedal before releasing the parking brake. Failure to do so will cause the vehicle to move as soon as you release the parking brake.



CAUTION:

The brake fluid may overflow.



9.Important additional information

1. Supplemental Restraint System (SRS)

Supplemental Restraint System (SRS) air bag component location ([Refer to 1-3.](#))

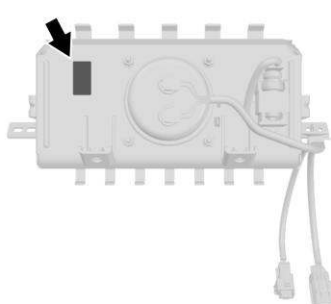
Caution Labels

The labels indicating the precautions for handling and maintenance of SRS air bags and seat belt with pre-tensioner are attached to the locations shown in the figure. If the label is damaged or is dirty, replace with the new label.

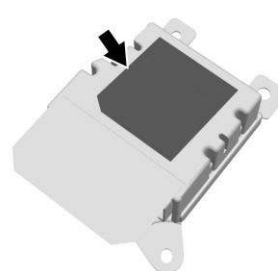
Driver's air bag module



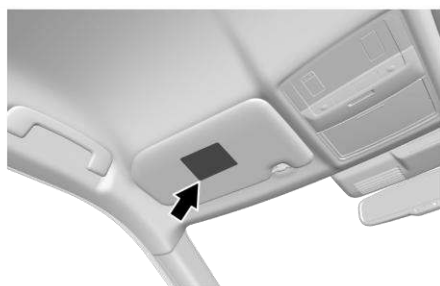
Passenger's (front) air bag



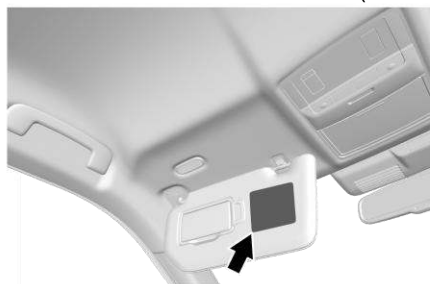
SRS-ECU



Sunvisor



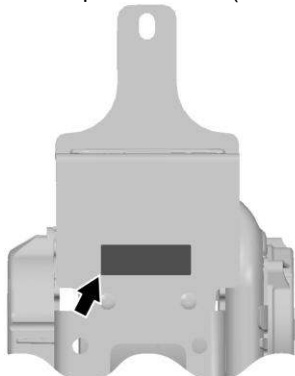
<Vehicles for USA and Canada (driver's side)>



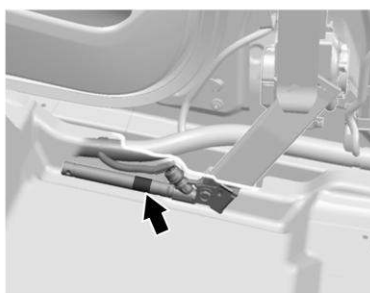
Seat belt pre-tensioner

(Driver's side and Passenger's side)

Second seat belt pre-tensioner (LH and RH)



Lap pre-tensioner

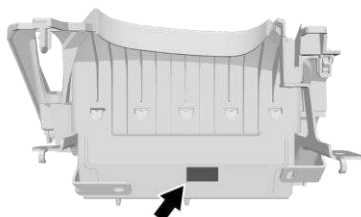


Side-airbag module

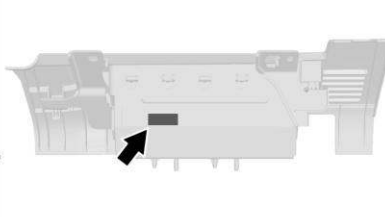
Center-airbag module



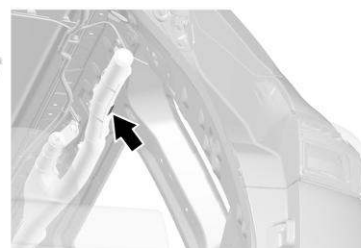
Driver's knee
air bag module



Passenger's knee
air bag module



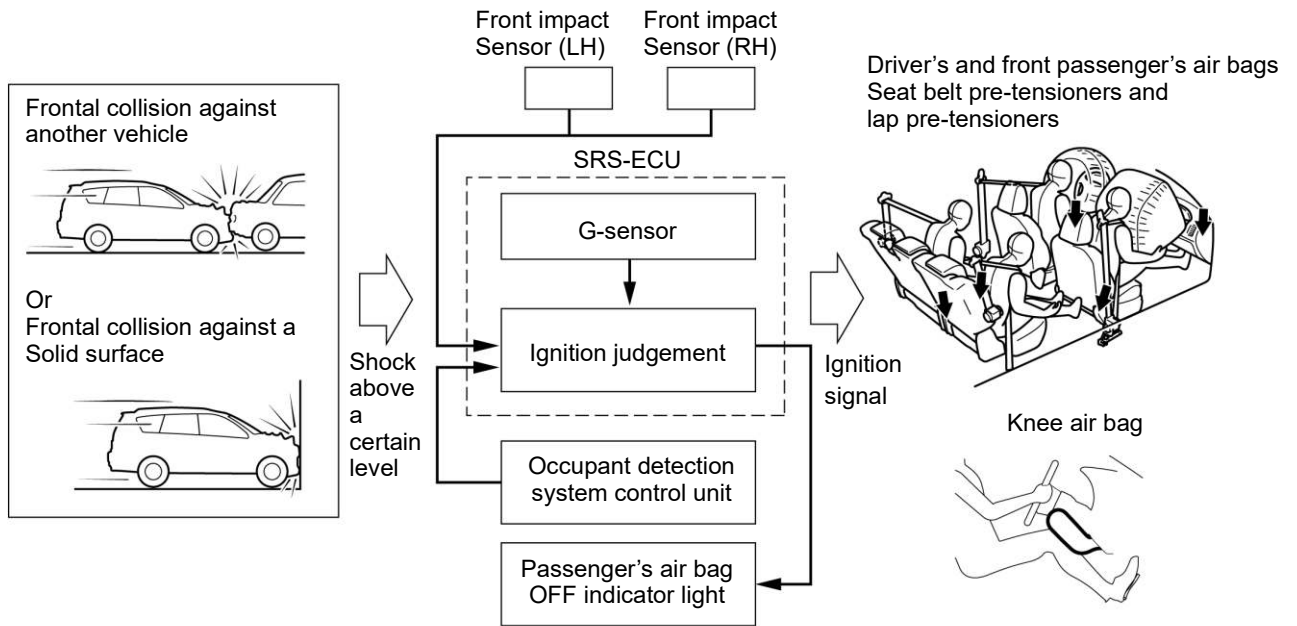
Curtain air bag module (LH and RH)



GM2W0035

System Operation

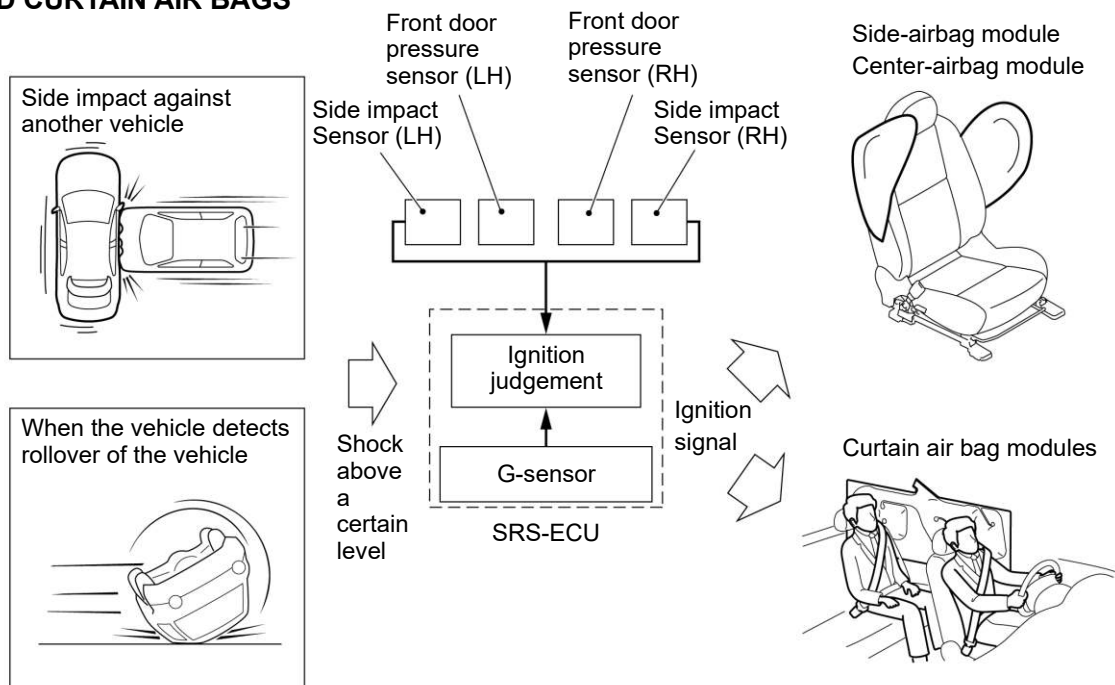
AIR BAGS, SEAT BELT PRE-TENSIONERS AND LAP PRE-TENSIONER



DE0072ENM0

SRS-ECU uses data of the front impact sensor (in engine compartment) and G-sensor (in SRS-ECU) to calculate collision severity during frontal collision. SRS-ECU judges necessity of air bag based on the calculated collision severity. In addition, the SRS-ECU also judges the operation of knee air bag, seat belt pre-tensioners and lap pre-tensioner after calculating collision severity. The occupant detection system control module recognizes the passenger's seat occupant. The occupant detection system control module identifies the occupant class based on the signals from occupant detection system sensor, and sends the signal to SRS-ECU.

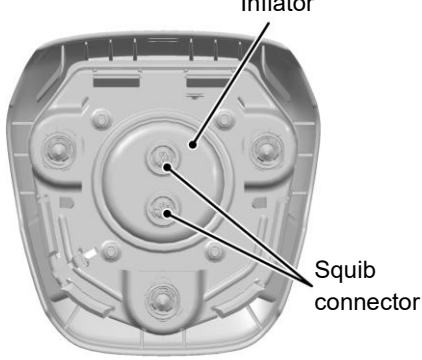
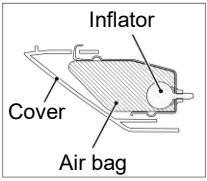
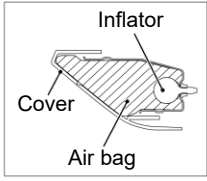
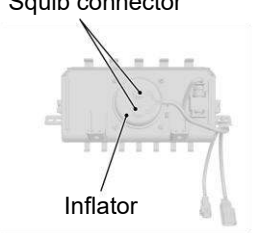
SIDE AND CURTAIN AIR BAGS

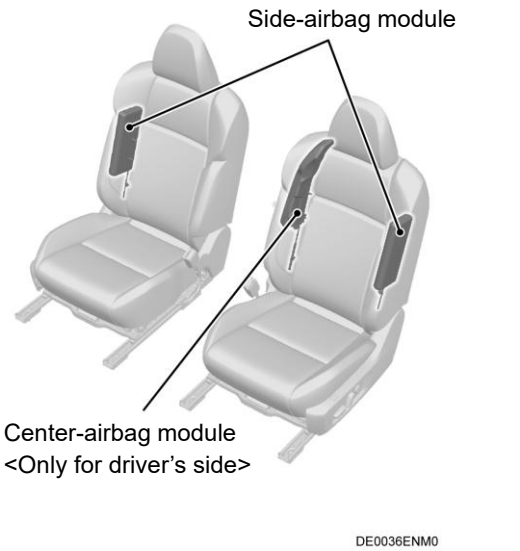
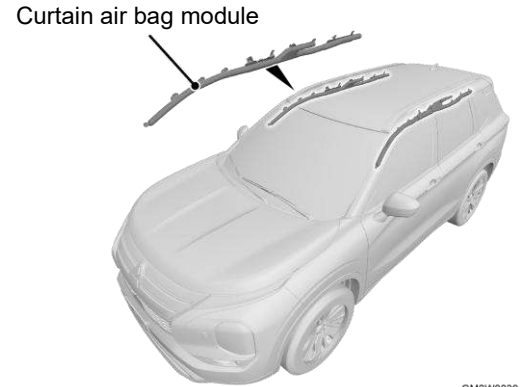
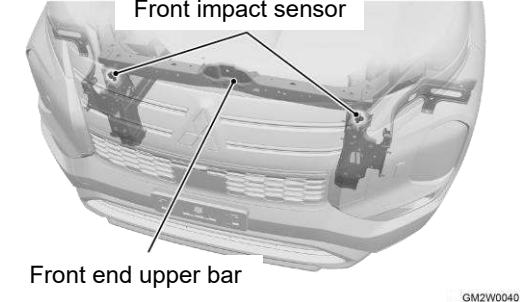


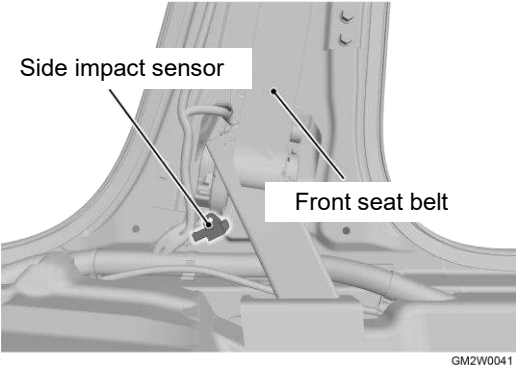
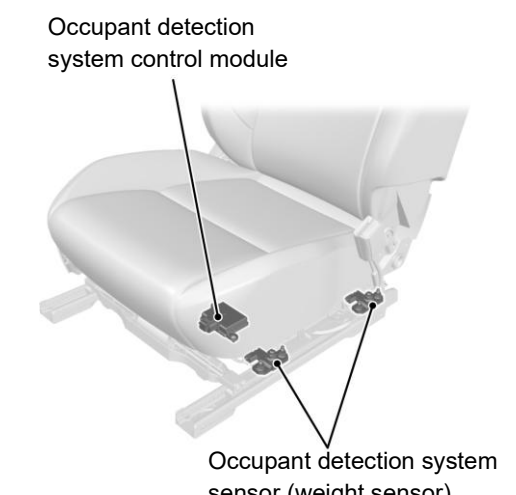
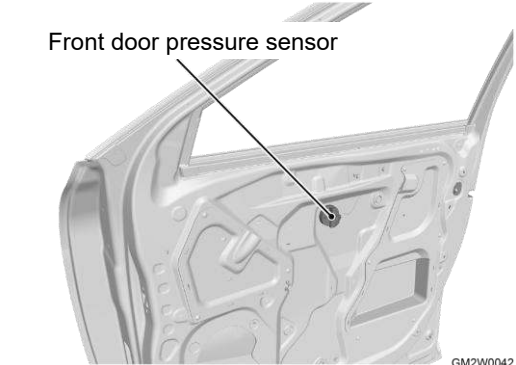
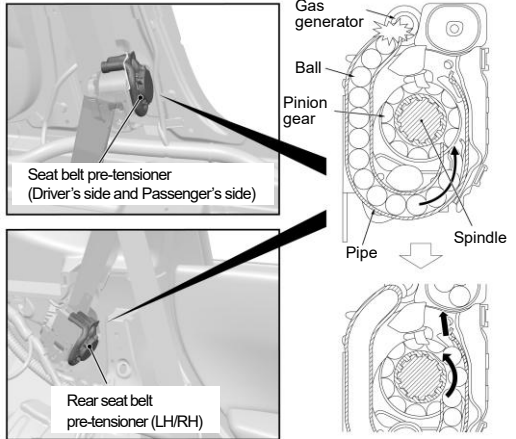
DE0032ENM0

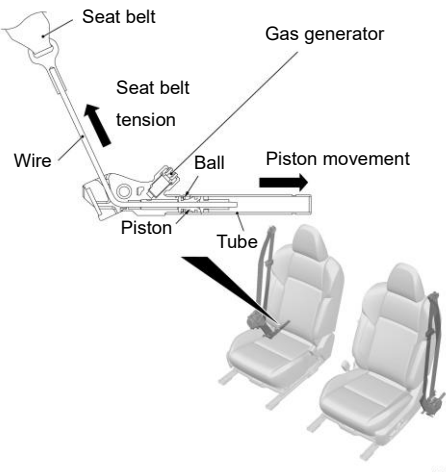
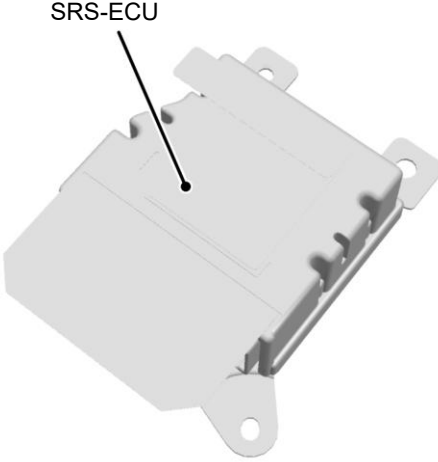
SRS-ECU uses data of the side impact sensor, front door pressure sensor, G-sensor (in SRS-ECU) and rollover sensor (in SRS-ECU) to calculate collision or rollover severity, during side collision or rollover. SRS-ECU judges necessity of side-airbag, center-airbag, curtain air bag, seat belt pre-tensioners and lap pre-tensioner based on the calculated collision or rollover severity.

System Construction

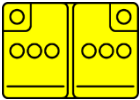
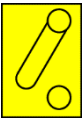
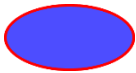
DRIVER'S AIR BAG MODULE	 Inflator Squib connector DE0033ENM0	The air bag module is equipped with an inflator that does not contain sodium azide. The air bag is made of nylon and is inflated by the gas from the inflator. Then, as the air bag contacts the passenger, the gas is discharged through two openings of backside of air bag, thus deflating the air bag to reduce the impact on the passenger. For the inflator, the gas which is harmless to the human body has been used.
DRIVER'S KNEE AIR BAG MODULE AND PASSENGER'S KNEE AIR BAG MODULE	<Driver's side>   Inflator Cover Air bag Driver's knee air bag module GM2W0036 <Passenger's side>   Inflator Cover Air bag Passenger's knee air bag module GM2W0037	The knee air bag module consists of a cover, an air bag, an inflator, and the fixing gear relating to those parts. The driver's knee air bag module is installed under the steering column and the passenger's knee air bag is installed under the glove box. Like the driver's and front passenger's air bags, the knee air bag is inflated upon a frontal collision for better protection of the driver and passenger. For the inflator, the gas which is harmless to the human body has been used.
PASSENGER'S (FRONT) AIR BAG MODULE	  Squib connector Inflator Passenger's (front) air bag module GM2W0038	The passenger's (front) air bag module consists of an air bag, and their fasteners. The air bag is made from nylon and inflates by the gas from the inflator. As a passenger is being pressed to the air bag, it deflates, discharging gas from vents at the side of the air bag to reduce the shock from the impact. For the inflator, the gas which is harmless to the human body has been used.

SIDE-AIRBAG MODULE AND CENTER-AIRBAG MODULE		The side-airbag module and center-airbag module consists of an air bag, air bag cover, inflator and their fasteners. The front side-airbag modules are installed in the outer side supports of the driver's and front passenger's seatbacks and the rear side-airbag modules are installed in the outer side of the second seat back assembly (LH and RH). Also, the center-airbag module is installed in the inner side support of the driver's seatback. The side-airbag module and the center-airbag module help protect the occupants regardless of the seat position and seatback angle. The air bags are compactly folded and stored under the cover. On the side of the air bag, there are holes through which gas is partially released to alleviate shock to the occupant when the air bag is deployed. For the inflator, the gas which is harmless to the human body has been used.
CURTAIN AIR BAG MODULE		The curtain air bag module consists of an air bag, an inflator, and the fixing gear relating to those parts, and is installed in the roof side sections (from the driver's and the passenger's front pillars to the quarter trim uppers). An inflator that does not contain sodium azide is used.
FRONT IMPACT SENSOR		The front impact sensor is installed on the front end upper bar in the engine room, and the analogue G-sensor is housed in the front impact sensor. The front impact sensor transmits the coded acceleration data to SRS-ECU. Based on the data, SRS-ECU determines the deployment stage of the front air bag, then energizes appropriate squib. SRS-ECU performs the diagnostics of the front impact sensor internal components. If a malfunction occurs, it sets the diagnosis trouble code.

SIDE IMPACT SENSOR	 Side impact sensor Front seat belt GM2W0041	The side impact sensors are installed to the lower part of the center pillars on both sides and lower part of quarter panel, and the analogue G-sensor is housed in the side impact sensor. The side impact sensor transmits the coded acceleration data to SRS-ECU. Based on the data, SRS-ECU determines the deployment of the side-airbags, center-airbag and curtain air bags, then energizes appropriate squib(s). SRS-ECU performs the diagnostics of the side impact sensor internal components. If a malfunction occurs, it sets the diagnosis trouble code.
OCCUPANT DETECTION SYSTEM CONTROL MODULE AND OCCUPANT DETECTION SYSTEM SENSOR	 Occupant detection system control module Occupant detection system sensor (weight sensor) DE0074ENM0	The occupant detection system sensor (weight sensor) is attached to the passenger's seat cushion frame. When the front passenger's seat is not occupied or the child is occupied, the passenger's airbag off indicator will stay on to show that the passenger's (front) air bag is not operational.
FRONT DOOR PRESSURE SENSOR	 Front door pressure sensor GM2W0042	The door pressure sensors are installed inside of the front door panels and the front door pressure sensor has a sensor that detects the internal pressure of the door in the event of a side collision The door pressure sensor transmits the coded acceleration data to SRS-ECU. Based on the data, SRS-ECU determines the deployment of the side-airbags, center-airbag and curtain air bags, then energizes appropriate squib(s).
SEAT BELT PRE-TENSIONER	 Seat belt pre-tensioner (Driver's side and Passenger's side) Rear seat belt pre-tensioner (LH/RH) Gas generator Ball Pinion gear Pipe Spindle DE0041ENM0	The seat belt incorporating the pre-tensioner automatically winds the seat belt upon front impact to reduce forward shifting of the passenger. The seat belt pre-tensioners of the driver's seat, front passenger's seat, and rear seat (LH, RH) ignite the gas generator and emit gas with the electric current supplied from the SRS-ECU. The gas pressure moves the ball in the pipe and the ball comes into contact with the recessed section of the pinion gear and the spindle connected to the pinion gear rotates. The belt is wound onto the rotating spindle.

LAP PRE-TENSIONER		The lap pre-tensioner automatically pulls in the seat belt upon a frontal collision to reduce the forward travel distance of occupants. Upon a collision, when the front impact sensor, attached to the front of the vehicle body, detects an impact that exceeds the permissible limit, the lap pre-tensioner ignites the gas generator with the signal from SRS-ECU, and the alternator emits the gas. By the gas pressure, the wire linked to the piston travels in the direction of movement, then the seat belt is pulled in. Subsequently, when a tension is applied to the seat belt by the forward travel of an occupant, balls are jammed between the piston and tube because of the wedge action, and the drawing position is maintained due to the locking.
SRS-ECU		In addition to the analogue G-sensor and safing G-sensor for the frontal collision, the SRS-ECU incorporates the analogue G-sensor and safing G-sensor for the side collision. In the event of a frontal collision, when the front impact sensor in the engine room and the analogue G-sensor for the frontal collision in the SRS-ECU detect the collision G at the same time, the SRS-ECU applies electrical current to the driver's seat/front passenger's seat air bag module, the driver's/passenger's knee air bag module, the front/rear seat belt pre-tensioner (squib) and the lap per-tensioners. Upon a side collision, when the lateral impact G is simultaneously detected by the analogue G-sensor of side impact sensor and the safing G-sensor for side collision in SRS-ECU, the SRS-ECU outputs the ignition signal to the side-airbag module (squib), center-airbag (squib), the curtain air bag module (squib), the front/rear seat belt pre-tensioner (squib) and the lap per-tensioners of the relevant side that is subjected to the impact. If a rollover occurs, the rollover sensor in the SRS-ECU detects it and the SRS-ECU energizes the curtain air bag module (squib), the front/rear seat belt pre-tensioner (squib) and the lap pre-tensioner. Also, SRS-ECU has the following functions that improve the system reliability. • Power supply back-up function (back-up condenser) in preparation for the power supply malfunction upon a collision • Booster function (DC-DC converter) in preparation for the battery voltage drop • Diagnosis function that performs a system check

10. Explanation of pictograms used

	12V starter battery		SRS unit		Keyless operation key distance
	Engine switch		Air bag inflator		Warning, Electricity
	Airbag		Seat belt pretensioner		Adjustment seat forward or backward
	Warning, low temperature		Adjustment seat height		Air-conditioning component
	Steering wheel height adjustment		Lifting point		Fuel tank
	Use ABC powder to extinguish the fire		Use water to extinguish the fire		Risk of damaging human health
	Risk of flammability		Risk of an explosion		Risk of corrosive material / substances

